

Asymptotically good construction of EAQECCs with controlled amounts of entanglement

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$C \subset \mathbf{F}_q^n$ with $C \supset C^\perp$ gives an $[[n, 2 \dim C - n, d_H(C \setminus C^\perp)]]_q$ QECC (quantum error-correcting code), encoding $(2 \dim C - n)$ qudits into n qudits and detecting up to $d_H(C \setminus C^\perp) - 1$ errors.

c pairs of maximally entangled states in $\mathbf{C}^q \otimes \mathbf{C}^q$ shared between an encoder and a decoder enable us to use any $C \subset \mathbf{F}_q^n$ with $c = \dim C^\perp - \dim C^\perp \cap C$ to construct entanglement-assisted (EA) QECC, with parameters denoted as $[[n, c + 2 \dim C - n, d_H(C \setminus C^\perp); c]]_q$.

Practical meaning of amount c of entanglement

Generation and preservation of shared entanglement between an encoder and a decoder is costly and currently almost impossible except for a short duration.

Amount $c = \dim C^\perp - \dim C^\perp \cap C$ should be smaller for quantum error correction.

An EAQECC is said to have maximal entanglement (ME) if its underlying linear code C is LCD $C \cap C^\perp = \{\mathbf{0}\}$, which gives the maximum possible value $c = \dim C^\perp$ for fixed dimensions of C and $C^\perp$.

The (Euclidean) hull of C is $C \cap C^\perp$ and its dimension is the hull dimension of C (and $C^\perp$).

Asymptotically good code constructions

A sequence of $[n_i, k_i, d_i]$ linear codes $C_i \subset \mathbf{F}_q^{n_i}$ is said to be asymptotically good if $n_i \rightarrow \infty$, $k_i/n_i \rightarrow R > 0$ and $d_i/n_i \rightarrow \delta > 0$ as $i \rightarrow \infty$.

We will review previous asymptotically good constructions of EAQECCs.

Known asymptotically good EAQECCs

1. F. R. F. Pereira, R. Pellikaan, G. G. La Guardia, and F. M. de Assis. Entanglement-assisted quantum codes from algebraic geometry codes. *IEEE Trans. Inform. Theory*, 67(11):7110–7120, Nov. 2021.
2. J. Fan, W. Cheng, G. Luo, Z. Li, and M. Cao. Entanglement-assisted concatenated quantum codes: Parameters and asymptotic performance, Jan. 2025. arXiv:2501.04921.

The previous asymptotic constructions yield maximal entanglement EAQECCs and the amount c of entanglement cannot be adjusted.

Construction of q -ary quantum codes from q^m -ary linear codes

$\{\alpha_1, \dots, \alpha_m\}, \{\beta_1, \dots, \beta_m\}$: Bases of $\mathbf{F}_{q^m}/\mathbf{F}_q$ such that $\text{Tr}_{q^m/q}[\alpha_i\beta_j] = \delta_{ij}$, which always exist.

$C \subset \mathbf{F}_{q^m}^n$ with $c = \dim C^\perp - \dim C^\perp \cap C$ and $k = c + 2 \dim_{\mathbf{F}_{q^m}} C - n$.
By expanding $C \times C \subset \mathbf{F}_{q^m}^{2n}$ to $C_X \times C_Z \subset \mathbf{F}_q^{2mn}$ by using two bases $\{\alpha_1, \dots, \alpha_m\}, \{\beta_1, \dots, \beta_m\}$, one can get an

$$[[mn, mk, \geq d_H(C \setminus C^\perp); mc]]_q$$

EAQECC, which follows from

A. Ashikhmin and E. Knill. Nonbinary quantum stabilizer codes.
IEEE Trans. Inform. Theory, 47(7):3065–3072, Nov. 2001.

Hull variation method and trivial construction 1

1. H. Chen. On the hull-variation problem of equivalent linear codes. *IEEE Trans. Inform. Theory*, 69(5):2911–2922, May 2023.
2. S. E. Anderson, E. Camps-Moreno, H. H. López, G. L. Matthews, D. Ruano and I. Soprunov, Relative hulls and quantum codes. *IEEE Trans. Inform. Theory*, 70(5):3190–3201, May 2024.

The hull dimension $\dim C \cap C^\perp$ can be decreased by an equivalent code to C , but increase of hull dimension seems impossible except in special cases. This means that the amount of entanglement can be increased but cannot be decreased in general.

1. H. Stichtenoth. Transitive and self-dual codes attaining the Tsfasman-Vladut-Zink bound. *IEEE Trans. Inform. Theory*, 52(5):2218–2224, May 2006.

One can construct asymptotically good EAQECCs Q_i with arbitrary amounts c of entanglement by using equivalent codes of asymptotically good **self-orthogonal linear codes** C_i after decreasing the hull dimension $\dim C \cap C^\perp$. (Use expansion of larger symbols to smaller symbols if necessary.)

We will show asymptotically good EAQECCs that are better than the above trivial combination of known results. One of key ideas is **Steane's enlargement construction**, which gave better asymptotic constructions of non-EA QECCs during 2000–2002.

Steane's enlargement construction (non-EA)

$\mathbf{F}_q$ -linear codes $C_2 \supset C_1 \supset C_1^\perp$ give an

$$[[n, k_1 + k_2 - n, \min\{d_1, \lceil (q+1)d_2/q \rceil\}]]_q$$

non-EA QECC, where $[n, k_i, d_i]$ are the parameter of C_i .

The number of information symbols is increased by $k_2 - k_1$.

1. A. M. Steane. Enlargement of Calderbank-Shor-Steane quantum codes. *IEEE Trans. Inform. Theory*, 45(7):2492–2495, Nov. 1999. (binary case)
2. S. Ling, J. Luo, and C. Xing. Generalization of Steane's enlargement construction of quantum codes and applications. *IEEE Trans. Inform. Theory*, 56(8):4080–4084, Aug. 2010. (general $\mathbf{F}_q$)

The conventional Steane's enlargement does not support EAQECCs. We will review an EA version.

Steane's enlargement construction with entanglement assistance

CSS Construction: C is an $[n, k]_q$ linear code with $c = \dim C^\perp - \dim(C \cap C^\perp)$. One can construct an EAQECC with parameters

$$[[n, c + 2k - n, d_H(C \setminus C^\perp); c]]_q.$$

Enlargement Construction: $C_t \subseteq C \cap C^\perp$ such that $\dim C_t \geq 2$, $C^\perp = C_r \oplus C_t$ (direct sum, i.e., $C_r \cap C_t = \{\mathbf{0}\}$). One can construct an EAQECC with parameters

$$[[n, c + 2k - n + \dim C_t, d'; c]]_q,$$

where

$$d' \geq \min \left\{ d_H(C), \left\lceil \frac{1+q}{q} d_H(C_r^\perp) \right\rceil \right\}$$

The number of information symbols is increased by $\dim C_t$.

C. Galindo, et al., Steane enlargement of entanglement-assisted quantum error-correcting codes. *Comp. Appl. Math.*, 42:275, Aug.

2023

Hull dimension adjustment and Steane's enlargement

- ▶ Equivalent codes with adjusted hull dimensions change $C_t \subseteq C \cap C^\perp$, $C^\perp = C_r \oplus C_t$, or $d_H(C_r^\perp)$.
- ▶ Use of the standard CSS construction degrades the performance (details shown at the end).

Asymptotically good self-orthogonal AG codes

$E_i/\mathbf{F}_{q^2}(z)$: tower of algebraic function fields in Stichtenoth (2006)
 $P_1, \dots, P_{n_i}$: all rational places of E_i such that $(z-1)(P_j) = 0$ for $j = 1, \dots, n_i$.

g_i : genus of E_i , satisfying $n_i/g_i \rightarrow q-1$ as $i \rightarrow \infty$.

There exists a positive divisor G_0 of E_i such that there exists a differential η with $(\eta) = 2G_0 - (P_1 + \dots + P_{n_i})$.

For a divisor G , $C(G) = \{(f(P_1), \dots, f(P_{n_i})) : f \in \mathcal{L}(G)\}$.

For any divisor G such that $P_j \notin \text{supp}(G)$, there exists

$\mathbf{a} \in (\mathbf{F}_{q^2} \setminus \{0\})^{n_i}$ independently of G such that

$$(\mathbf{a} \cdot C(G_0 + G))^\perp = \mathbf{a} \cdot C(G_0 - G)$$

where $\cdot$ means component-wise product (Schur product).

One can construct asymptotically good AG codes with arbitrary hull dimensions, by using a suitable G .

Choice of C , C_r and C_t in Steane's enlargement

Let P_0 and P_∞ be the unique zero and the unique pole of z in $E_i/\mathbf{F}_{q^2}$.

$$\begin{aligned}C &= \mathbf{a} \cdot C(G_0 + \ell_\infty P_\infty - \ell_0 P_0), \\C^\perp &= \mathbf{a} \cdot C(G_0 - \ell_\infty P_\infty + \ell_0 P_0), \\C \cap C^\perp &= \mathbf{a} \cdot C(G_0 - \ell_\infty P_\infty - \ell_0 P_0), \\C_r &= \mathbf{a} \cdot C(G_0 - \ell_r P_\infty + \ell_0 P_0).\end{aligned}$$

Let $f_j \in \mathcal{L}(G_0 - (\ell_r - j)P_\infty - \ell_0 P_0) \setminus \mathcal{L}(G_0 - (\ell_r - j + 1)P_\infty - \ell_0 P_0)$ for $j = 1, \dots, \ell_r - \ell_\infty$, and let C'_t be a linear space spanned by $(f_j(P_1), \dots, f_j(P_{n_i}))$ for $j = 1, \dots, \ell_r - \ell_\infty$, and $C_t = \mathbf{a} \cdot C'_t$.

EAQECC parameters 1

$[[n_i, k_i, d_i; c_i]]_{q^2}$ with

$$c_i = 2\ell_0,$$

$$k_i = \ell_r + \ell_\infty,$$

$$d_i \geq \min \left\{ 1 + n_i/2 - g_i - \ell_\infty + \ell_0, \right. \\ \left. \left\lceil \frac{q^2 + 1}{q^2} (1 + n_i/2 - g_i - \ell_r + \ell_0) \right\rceil \right\}$$

Information rate satisfies

$$\frac{k_i}{n_i} \rightarrow 1 + \theta - 2\gamma - \frac{2q^2 + 1}{q^2 + 1} \delta,$$

as $i \rightarrow \infty$, where $\gamma = g_i/n_i$, $\theta = c_i/n_i$ and δ is (the lower bound on) relative Hamming distance.

EAQECC parameters 2

Information rate satisfies

$$\frac{k_i}{n_i} \rightarrow 1 + \theta - 2\gamma - \frac{2q^2 + 1}{q^2 + 1}\delta,$$

as $i \rightarrow \infty$, where $\gamma = g_i/n_i$, $\theta = c_i/n_i$.

Straightforward combination of the CSS construction, Stichtenoth's tower of function fields, and hull dimension adjustment gives information rate

$$1 + \theta - 2\gamma - 2\delta$$

which is smaller than the proposed construction by $\frac{1}{q^2+1}\delta$.

To construct even better codes, we should use Steane's enlargement after expanding linear codes over a larger field (whose choice depends on δ) to a smaller target field, which improves the ratio $\frac{2q^2+1}{q^2+1}$ by decreasing q .